

# Tuan Nguyen

PhD Student in Computer Science

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## Education

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| <b>PhD</b> | <b>University of Warwick</b> , Computer Science <ul style="list-style-type: none"> <li>Fully funded by an EPSRC Doctoral Scholarship</li> <li>Supervisor: <a href="#">Prof. Long Tran-Thanh</a></li> <li>Research focus: AI safety and alignment for large language models</li> </ul>                                                    | Coventry, UK<br>Oct 2025 – present  |
| <b>MSc</b> | <b>Toyo University</b> , Computer Science <ul style="list-style-type: none"> <li>Supervisors: <a href="#">Prof. Shugo Nakamura</a> (main) and <a href="#">Prof. Hirotada Honda</a></li> <li>Research focus: graph neural networks, Transformers and applications of machine learning</li> <li>Excellent Master's Thesis Award</li> </ul> | Tokyo, Japan<br>Apr 2022 – Apr 2024 |
| <b>BSc</b> | <b>Toyo University</b> , Computer Science and Engineering <ul style="list-style-type: none"> <li>Full-ride scholarship</li> <li>CGPA: 4.17/4.3 (summa cum laude)</li> </ul>                                                                                                                                                              | Tokyo, Japan<br>Apr 2018 – Mar 2022 |

## Experience

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| <b>FPT AI Center</b> , Machine Learning Researcher <ul style="list-style-type: none"> <li>Designed and developed features for a <b>large-scale traffic flow</b> simulation project, using Physics-Informed Neural Networks (PINNs) to address complex urban planning challenges.</li> <li>Inspired by the <b>Kuramoto model</b> from physics, developed a framework that bridges computational physics and deep learning to mitigate <b>oversmoothing in graph neural networks</b>, achieving state-of-the-art results; presented at <b>AISTATS 2024</b>.</li> <li>Optimised algorithms for a long-term <b>demand forecasting</b> project for a leading pharmaceutical company in Vietnam, leading to a <b>10% reduction in stockouts</b> and significant revenue growth.</li> </ul> | Tokyo, Japan<br>Jan 2022 – Sept 2025<br>3 years 9 months  |
| <b>Aimesoft JSC</b> , Machine Learning Researcher <ul style="list-style-type: none"> <li>Improved multimodal fake-news detection by 4% over baselines using <b>BERT</b> and <b>VGG19</b>.</li> <li>Implemented <b>BM25</b> retrieval with <b>BERT</b> for Vietnamese legal text processing.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Remote, Vietnam<br>Apr 2020 – May 2021<br>1 year 2 months |
| <b>Tokyo Techies</b> , Machine Learning Intern <ul style="list-style-type: none"> <li>Deployed <b>deep learning on edge devices</b> for facial recognition in a productivity project, reaching 80% accuracy even with masks and cutting day-long video inference time to 5 hours.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Tokyo, Japan<br>Feb 2020 – Aug 2021<br>1 year 7 months    |

## Selected Publications

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- **Tuan Nguyen**, Long Tran-Thanh. [Safety Game: Inference-Time Alignment of Black-Box LLMs via Constrained Optimization](#). *International Conference on Machine Learning (ICML)*, 2026.
- Kiet Huynh, Minh Dao, **Tuan Nguyen**, Chi-Nguyen Tran, Phu-Hoa Pham, Quy Nguyen, Anh Han, Long Tran-Thanh. [Training-Free Cultural Alignment of Large Language Models via Persona Disagreement](#). *Advances in Neural Information Processing Systems (NeurIPS)*, 2026.
- **Tuan Nguyen**, Hirotada Honda, Takashi Sano, Vinh Nguyen, Shugo Nakamura, Tan Nguyen. [From Coupled Oscillators to Graph Neural Networks: Reducing Over-smoothing via a Kuramoto Model-based Approach](#). *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2024.
- **Tuan Nguyen**, Minh Pham. [Multimodal Fusion with BERT and Attention Mechanism for Fake News Detection](#). *RIVF International Conference on Computing and Communication Technologies*, 2021.
- **Tuan Nguyen**, Minh Pham. [ReINTEL Challenge 2020: A Multimodal Ensemble Model for Detecting Unreliable Information on Vietnamese SNS](#). *Proceedings of the 7th International Workshop on Vietnamese Language and Speech Processing (VLSP)*, 2020.

## Awards and Scholarships

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- **EPSRC Doctoral Scholarship**, University of Warwick: full tuition and a UKRI stipend of £20,780 per year for 3.5 years, for exceptional academic achievement and research potential.
- **Toyo University Full-ride Scholarship**: full tuition and an annual stipend of nearly USD 12,000 for 4 years, for outstanding academic performance.
- **Best Presentation Award**, 15th IEEE RIVF International Conference, 2021.
- **Second Prize**, Automated Legal Question Answering Competition (ALQAC), 2021.
- **Outstanding Solution**, Reliable Intelligence Identification on Vietnamese SNSs (ReINTEL) Competition, 2020.
- Shopee Code League 2020: ranked **29/356** teams in Sentiment Analysis and **97/823** teams in Product Image Classification.
- SIGNATE Student Cup 2019, House Price Prediction: ranked **39/300** teams.
- **Bronze Prize**, HSGS Olympiad 2015 (nationwide competitive programming for specialised high school students).
- **Silver Prize**, Competitive Programming, Competition for Coastal and Northern Region Students in Vietnam, 2015.
- **Third Prize**, Vietnam National University-Level Competitive Programming Competition, 2015 (earned exemption from the national university admission process).